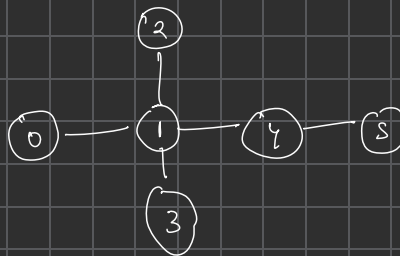
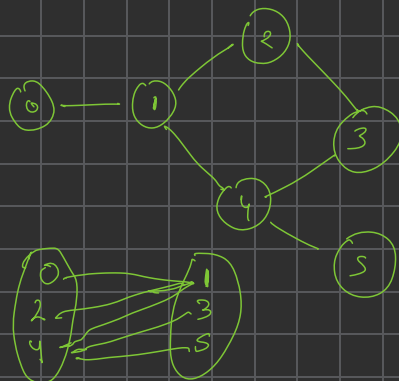
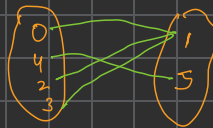




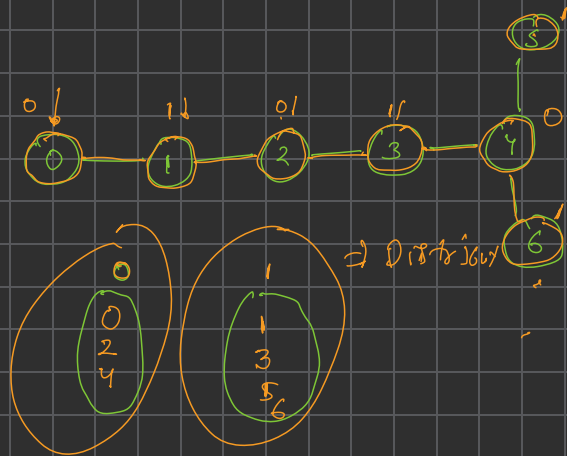
[Bipartite graph]:

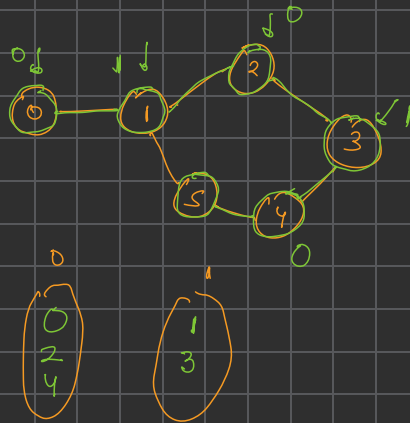


Node is 2 group
distribute,
Node in
a same should
not be connected
through a
edge

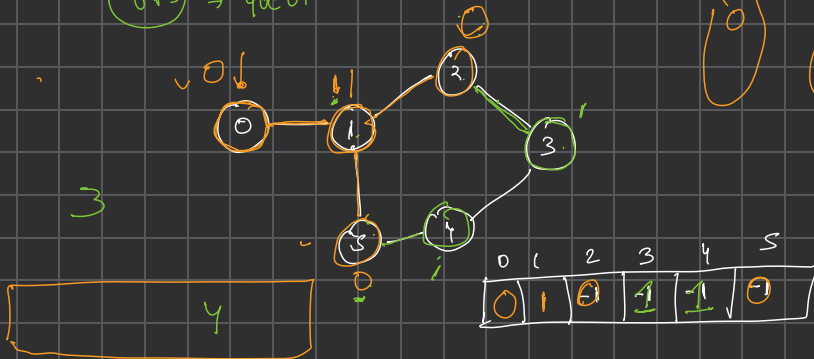
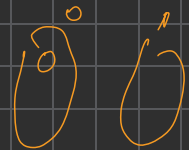


Bipartite
graph





BFS $\Rightarrow$ queue

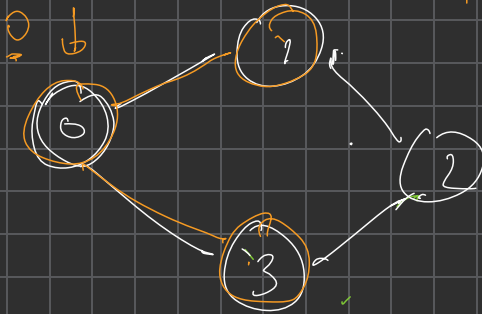


$$\text{color}[i] = (\text{color}[0] + 1) \% 2$$

7 0 0

$$(0+1) \% 2$$

$$(1+1) \% 2$$



0	1	2	3
0	1	-1	1

↑ color

1 3

return 1

Bipartite

- ① create a color vector of size V and assign all values -1
- ② push 0 into queue and assign color as 0 to it
- ③ while queue is not empty

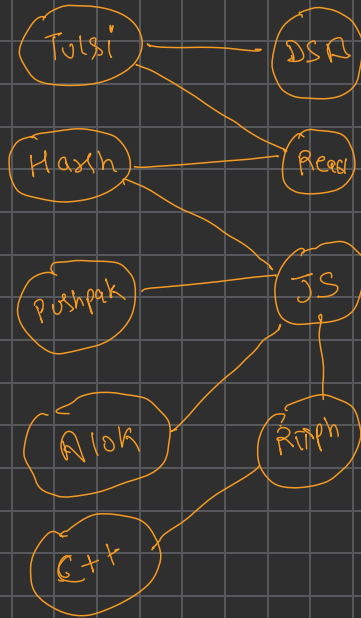
① Get the first node

② Look at its all neighbours

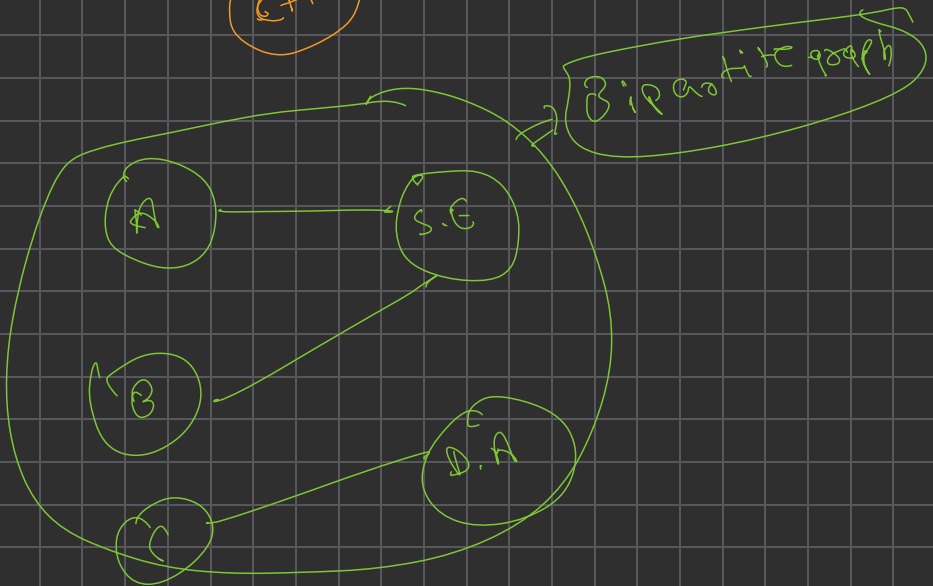
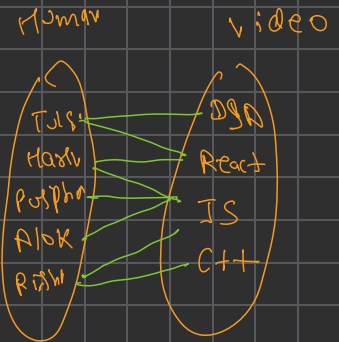
→ if neighb color is -1 (parent color)
 $\text{color}[\text{neib}] = (\text{color}[\text{node}] + 1) \% 2$;
 push neib into queue

→ if neighb color == node
 return 0

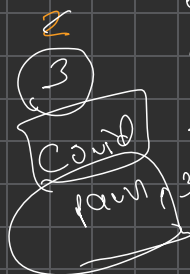
1)



Youtube
recommended



Timer = 0
x



	0	1	2	3	4
0	2	2	0	0	2
1	2	0	2	0	2
2	2	0	2	2	2
3	2	0	2	0	0

- 0 - Empty
- 1 - In affected path
- 2 - Covid porta

	0	1	2	3	4
0	2	1	0	0	1
1	1	0	1	0	2
2	1	0	2	1	1
3	1	0	1	0	0

2	0	0	0	0
2	0	0	0	0
2	2	2	2	0
0	0	0	0	0

1st row — 2nd row

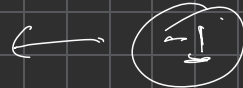
← ↑ → ↓

2nd row

2	0	0	2
2	0	2	2
0	0	0	0
0	0	0	0

1st row

2nd row



	0	1	2	3	4
0	0	2 → 2	0	0	0
1	2	2	0	0	0
2	2	2	0	2	2
3	2	2	0	2	0

Timer = 0

Timer = 1

2
3
4

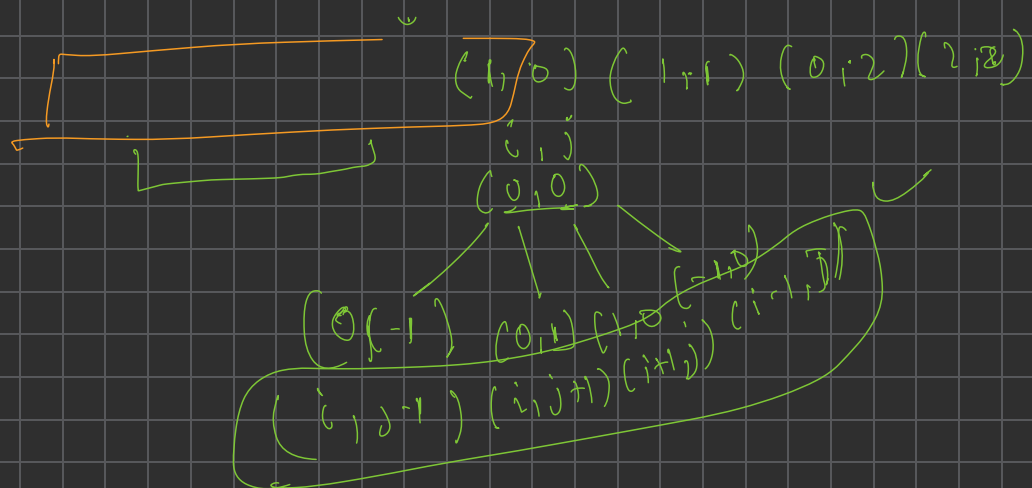


$(y-1) \Rightarrow (3) \text{ answer}$

hospital [0] [1]
 ↗ cross

Timer = 0 + [2] ✓
 Current patient = 4

	0	1	2	3
0	(2)	0	2	0
1	2	2	(2)	0
2	0	1	2	0
3	0	1	1	0



2 < 0

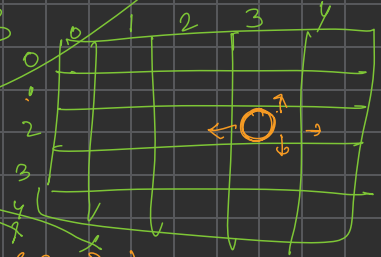
(3, -1)



2 > 0
2 < 5
2 > 0
2 < 3

out of index

(i, j) = (2, 3)



(2, 2) (2, 4) (1, 3) (3, 3)
Left Right Up down

int row[4] = {0, 0, -1, 1}
int col[4] = {-1, 1, 0, 0}

Loop

for (int k = 0; k < 4; k++)
{
if (hospital[i + row[k]][j + col[k]])

}

int row[4] = {0, 0, 1, -1}
int col[4] = {1, -1, 0, 0}

